

Data Science: Basic to Advanced Knowledge Base

Text-only RAG training document | Fundamentals, statistics, machine learning, deep learning, NLP, RAG, MLOps, and advanced topics

This text-only document is organized as a retrieval-oriented knowledge base for RAG systems. Sections use explicit concepts, definitions, workflows, formulas, examples, and terminology so that retrieved chunks can be understood independently.

Educational note: content is general information. It is not a substitute for professional advice, and current facts such as financial regulations, prices, or broker conditions should be verified from current authoritative sources.

1. Data Science Fundamentals

What is Data Science?

Data science is an interdisciplinary field that combines statistics, mathematics, programming, data engineering, machine learning, and domain knowledge to extract useful information from data and support decisions.

Data, Information, and Knowledge

Data consists of recorded observations. Information is organized data that answers a question. Knowledge connects information to interpretation, action, or domain understanding.

Data Science Lifecycle

- Define the problem, users, constraints, and success criteria.
- Collect or access relevant data.
- Profile and validate data quality.
- Clean, transform, and integrate data.
- Explore patterns using statistics and analysis.
- Engineer useful features.
- Train, validate, and test models when prediction is required.
- Evaluate with metrics aligned to the real problem.
- Deploy when appropriate.
- Monitor data, model behavior, errors, and business impact.
- Document assumptions, limitations, and results.

2. Data Types, Sources, and Quality

Data Types

Structured data follows a defined schema, such as relational tables. Semi-structured data has organizational tags or keys, such as JSON and XML. Unstructured data includes documents, text, images, audio, and video.

Common Sources

- Relational databases and data warehouses.
- NoSQL databases.
- CSV, Excel, JSON, XML, and Parquet files.
- APIs and web services.
- Application logs, sensors, and event streams.
- Cloud storage and enterprise systems.

Data Quality

Important dimensions include accuracy, completeness, consistency, timeliness, validity, uniqueness, and integrity. A model cannot reliably compensate for systematically poor source data.

3. Python and the Data Science Ecosystem

Python

Python is widely used for data science because of its readable syntax and ecosystem for numerical computing, data manipulation, statistics, machine learning, deep learning, and deployment.

Core Python

Important foundations include variables, data types, strings, lists, tuples, sets, dictionaries, conditions, loops, functions, modules, exceptions, file handling, classes, virtual environments, and dependency management.

Major Libraries

- NumPy: arrays and numerical computation.
- pandas: tabular data loading, cleaning, grouping, joining, and analysis.
- SciPy: scientific and statistical routines.
- Matplotlib: general-purpose visualization.
- scikit-learn: classical machine learning and evaluation.
- XGBoost, LightGBM, and CatBoost: gradient-boosted trees.
- PyTorch and TensorFlow: deep learning.

4. SQL and Databases

SQL

SQL is used to query and manipulate relational data. Core operations include SELECT, WHERE, GROUP BY, HAVING, ORDER BY, JOIN, INSERT, UPDATE, and DELETE.

Joins

INNER JOIN returns matching rows. LEFT JOIN preserves every row from the left table and adds matching rows from the right table. Other join types depend on database support and use case.

Aggregation

COUNT, SUM, AVG, MIN, and MAX summarize data. GROUP BY defines the grouping level and HAVING filters groups after aggregation.

Window Functions

Window functions calculate values across related rows without collapsing the result set. Examples include ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD, and running totals.

5. Statistics and Probability

Descriptive Statistics

Mean is the arithmetic average. Median is the middle ordered value. Mode is the most frequent value. Variance measures average squared deviation from the mean; standard deviation is its square root.

Probability

Probability represents uncertainty. Conditional probability describes an event given another event. Bayes' theorem connects conditional probabilities and is widely used in probabilistic inference.

Distributions

Common distributions include Bernoulli, binomial, categorical, Poisson, uniform, normal, exponential, and log-normal distributions. Choice depends on the data-generating process.

Sampling and Estimation

A sample is a subset of a population. Sampling design affects bias and uncertainty. Confidence intervals express uncertainty under a specified statistical procedure.

Hypothesis Testing

A hypothesis test compares observed data with a null hypothesis under defined assumptions. A p-value describes compatibility of the observed result with the null model; it is not the probability that the null hypothesis is true.

6. Data Cleaning and Exploratory Data Analysis

Missing Values

Missing data can be removed, imputed, or modeled depending on the amount, mechanism, and meaning of the missingness. Common approaches include median, mode, model-based, and domain-specific imputation.

Outliers

An outlier is unusually distant from other observations. It may be an error, rare legitimate event, fraud signal, sensor problem, or important edge case. Outliers should not be removed automatically.

Encoding

Categorical variables can use one-hot, ordinal, target, or other encodings. The choice should match variable meaning and model behavior.

Scaling

Standardization commonly creates zero-mean, unit-variance features. Min-max scaling maps values to a selected interval. Tree-based models are generally less sensitive to scaling than distance-based methods.

EDA

Exploratory analysis examines distributions, missingness, outliers, relationships, category frequencies, and suspicious records before modeling.

7. Machine Learning Fundamentals

Machine Learning

Machine learning fits patterns from data to make predictions, classifications, rankings, recommendations, or decisions.

Supervised Learning

Supervised learning uses labeled examples. Regression predicts numeric values; classification predicts categories.

Unsupervised Learning

Unsupervised learning works without a target label. Clustering discovers groups and dimensionality reduction creates lower-dimensional representations.

Reinforcement Learning

Reinforcement learning models an agent interacting with an environment. The agent selects actions and receives rewards or penalties with the objective of maximizing cumulative reward.

Train, Validation, and Test

Training data fits parameters. Validation data supports model selection and tuning. Test data estimates performance on unseen examples. Cross-validation provides repeated train-validation splits when appropriate.

Overfitting and Underfitting

Overfitting captures training-specific patterns that fail to generalize. Underfitting occurs when a model is too simple. Regularization, suitable complexity, better features, more data, and sound validation can help.

8. Classical Machine Learning Algorithms

Linear Regression

Linear regression models a numeric target as a linear combination of inputs. Ridge and lasso add regularization.

Logistic Regression

Logistic regression models class probabilities and is a strong, interpretable baseline for many classification problems.

Decision Trees

Decision trees recursively split data using feature-based rules. They are interpretable but can overfit without constraints.

Random Forest

Random forest combines many randomized decision trees. Aggregation can reduce variance and provide a strong baseline for tabular data.

Gradient Boosting

Gradient boosting builds learners sequentially so later learners focus on previous errors. XGBoost, LightGBM, and CatBoost are popular implementations.

Support Vector Machines

SVMs seek separating boundaries with a large margin. Kernels can represent nonlinear boundaries. Scaling is often important.

K-Nearest Neighbors

KNN predicts from nearby training observations. It is simple but sensitive to feature scale and computational cost at large dataset sizes.

Naive Bayes

Naive Bayes applies Bayes' theorem with a conditional-independence assumption and is often effective for text classification.

9. Model Evaluation

Regression Metrics

MAE is mean absolute error. MSE is mean squared error. RMSE is the square root of MSE and uses the target's units. R-squared describes explained variance under its standard definition.

Classification Metrics

Accuracy is the fraction of correct predictions. Precision measures correctness among predicted positives. Recall measures detection among actual positives. F1 is the harmonic mean of precision and recall. ROC-AUC evaluates ranking across thresholds; precision-recall analysis can be more informative for highly imbalanced tasks.

Confusion Matrix

A confusion matrix counts true positives, true negatives, false positives, and false negatives. It connects model errors to practical consequences.

Cross-Validation

K-fold cross-validation repeatedly trains and validates across folds. Time-dependent problems should use time-aware splits when future information must not leak into training.

10. Feature Engineering and Leakage

Feature engineering transforms raw data into representations useful to models. Examples include date parts, ratios, interactions, lag features, rolling statistics, and domain-specific transformations.

Feature selection can remove noisy or redundant variables. Filter methods use statistics, wrapper methods evaluate subsets with a model, and embedded methods perform selection during training.

Data leakage occurs when information unavailable at prediction time enters features or preprocessing. Leakage can create unrealistically high validation performance.

11. Unsupervised Learning

K-Means

K-means partitions observations into a chosen number of clusters by minimizing within-cluster squared distances to cluster centers.

Hierarchical Clustering

Hierarchical clustering creates nested groups. Agglomerative methods begin with individual observations and repeatedly merge clusters.

DBSCAN

DBSCAN identifies dense regions and noise points without requiring a predefined number of clusters. Its parameters depend on scale and density.

PCA

Principal Component Analysis transforms correlated features into orthogonal components ordered by explained variance. It can reduce dimensionality but may reduce interpretability.

12. Deep Learning

Neural Networks

A neural network contains layers of units. Each unit computes a weighted sum plus bias and applies an activation function. Training adjusts parameters to minimize a loss function.

Activations

ReLU is common in hidden layers. Sigmoid maps values to 0 through 1 and is common for binary outputs. Softmax converts multiple logits into a probability distribution.

Backpropagation

Backpropagation calculates gradients using the chain rule. Optimizers such as SGD and Adam update parameters using gradients.

CNN

Convolutional neural networks learn spatial patterns and are widely used for image classification, detection, and segmentation.

RNN, LSTM, and GRU

Recurrent architectures process sequences. LSTM and GRU use mechanisms designed to preserve useful information over longer sequences.

Transformers

Transformers use attention to model relationships among tokens or sequence elements and are foundational to modern language, vision, and multimodal models.

13. NLP and Computer Vision

Natural Language Processing

NLP includes tokenization, text classification, named entity recognition, sentiment analysis, summarization, translation, information extraction, and question answering.

Bag-of-words and TF-IDF are traditional representations. Embeddings provide dense numerical representations, while contextual embeddings represent tokens using surrounding context.

Computer Vision

Vision tasks include image classification, object detection, and segmentation. Evaluation may use accuracy, precision, recall, intersection-over-union, or mean average precision depending on the task.

14. Time Series and Forecasting

Time-series data is ordered by time. Important concepts include trend, seasonality, cycles, autocorrelation, stationarity, lags, rolling statistics, and forecast horizon.

Methods include moving averages, exponential smoothing, ARIMA-family models, tree-based models with lag features, and neural networks. Validation should preserve temporal order.

15. Recommender Systems

Recommendation systems estimate which items may be relevant to users. Collaborative filtering uses interaction patterns. Content-based systems use item attributes. Hybrid systems combine multiple signals.

Evaluation can use precision at k, recall at k, mean reciprocal rank, normalized discounted cumulative gain, and online business measures such as engagement or conversion.

16. Generative AI, Embeddings, and RAG

Generative AI

Generative AI models create new content such as text, code, images, audio, or multimodal outputs from learned representations and prompts.

Embeddings

An embedding is a numerical vector designed to represent semantic or structural relationships. Similarity can be measured using metrics such as cosine similarity.

Retrieval-Augmented Generation

RAG combines retrieval from an external knowledge base with language-model generation. A typical pipeline ingests documents, extracts text, chunks content, creates embeddings, stores vectors, retrieves relevant chunks, and supplies them as context to a generator.

RAG Quality

Important dimensions include retrieval recall, retrieval precision, context relevance, answer faithfulness, latency, cost, and robustness. Chunking, metadata, query rewriting, hybrid search, reranking, and citation strategies can affect performance.

Vector Databases

Vector databases store embeddings and support similarity search using indexes, metadata filters, approximate nearest-neighbor methods, and sometimes hybrid retrieval.

17. MLOps and Production

MLOps applies software engineering, data engineering, and operations practices to machine learning systems. It addresses reproducibility, versioning, testing, deployment, monitoring, governance, and lifecycle management.

- Version code, data definitions, configurations, and model artifacts.
- Automate validation and training where appropriate.
- Package dependencies reproducibly.
- Deploy through batch jobs, APIs, services, or embedded inference.
- Monitor latency, errors, drift, and business metrics.
- Maintain rollback and retraining procedures.

Models can be served through REST APIs, batch inference, streaming, or embedded applications depending on latency, throughput, reliability, cost, and privacy requirements.

18. Responsible and Advanced Data Science

Responsible data science considers privacy, fairness, transparency, security, accountability, and appropriate use. Bias can enter through collection, labeling, sampling, measurement, features, modeling, or deployment.

Security risks include unauthorized access, data leakage, insecure endpoints, malicious inputs, prompt injection in AI systems, and supply-chain vulnerabilities.

Causal Inference

Causal inference studies effects of interventions rather than simple association. Concepts include randomized experiments, confounding, treatment effects, causal graphs, and counterfactuals.

Bayesian Modeling

Bayesian methods represent uncertainty with probability distributions over parameters or hypotheses and update prior beliefs with observed data.

Survival Analysis

Survival analysis models time-to-event outcomes while handling censoring. Common concepts include survival functions, hazard functions, Kaplan-Meier estimation, and proportional hazards.

Anomaly Detection

Anomaly detection identifies observations that differ from expected patterns using statistical rules, isolation forests, one-class models, density methods, or autoencoders.

Graph Data Science

Graph methods model entities as nodes and relationships as edges. Applications include fraud detection, recommendations, knowledge graphs, and link prediction.

19. End-to-End Project Architecture

A production data science project can contain data ingestion, validation, storage, transformation, exploratory analysis, feature engineering, training, evaluation, model registry, deployment, monitoring, and documentation.

A strong project defines the problem, target, prediction point, data sources, split strategy, baseline, metrics, error analysis, limitations, and reproducibility instructions.

For RAG, preserve source metadata during ingestion. Chunking should balance specificity with context. Evaluate retrieval separately from generation so failures can be diagnosed at the correct stage.

Glossary

Data Science: Interdisciplinary use of data, statistics, programming, modeling, and domain knowledge.

Dataset: Collection of related observations or records.

Feature: Input variable used by a model.

Target: Value a supervised model is trained to predict.

Model: Fitted computational representation used for prediction or inference.

Overfitting: Learning patterns that do not generalize beyond training data.

Regularization: Technique that discourages overly complex model solutions.

Data Leakage: Use of information during training that would not be available at real prediction time.

Embedding: Numerical vector representation of an object such as text.

RAG: Retrieval-Augmented Generation using retrieved external context.

MLOps: Practices for reliable development, deployment, monitoring, and maintenance of ML systems.